

SARDAR VALLABHBHAI NATIONAL INSTITUTE OF TECHNOLOGY

Big Data analytics (AI358)

Lab Assignment - 7

Q1. Scenario: You have a massive dataset of global weather readings. Querying the entire table for a specific country is too slow.

- **Task:** Create an **External Table** for weather data, then create a **Partitioned Table** based on country and year.
- **Action:** 1. Load data from the external table into the partitioned table using dynamic partitioning. 2. Compare the execution time (or the number of HDFS files scanned) between a SELECT statement on the non-partitioned table vs. the partitioned table.
- **Key Concept:** SET hive.exec.dynamic.partition.mode=nonstrict; and performance tuning.

Q2. Scenario: A web server outputs logs in JSON format where one field, attributes, contains a map of key-value pairs (e.g., {"browser": "Chrome", "os": "Windows"}).

- **Task:** Create a Hive table using the **JSON SerDe** to parse this data.
- **Action:** 1. Use the EXPLODE function and LATERAL VIEW to turn the attributes map into individual rows. 2. Write a query to find the most common browser used by visitors.
- **Key Concept:** Handling non-tabular data and UDTFs (User-Defined Table-Generating Functions).

Q3. Scenario: You need to understand what happens to data when a table is dropped.

- **Task:** Create two tables—one **Internal (Managed)** and one **External**—pointing to the same HDFS directory.
- **Action:** 1. Drop both tables using the DROP TABLE command. 2. Check the HDFS file system using dfs -ls to see which data remains and which is deleted.
- **Key Concept:** Metadata vs. Actual Data management.

Q4. Scenario: You have a user table with millions of rows, and you want to join it with a transaction table. To optimize the join, you decide to bucket the data.

- **Task:** Create a table bucketed by user_id into 4 buckets.
- **Action:** 1. Use the TABLESAMPLE clause to extract a 25% random sample of users from a specific bucket. 2. Explain why bucketing is preferred over partitioning for high-cardinality columns like user_id.
- **Key Concept:** Data distribution and efficient sampling for large-scale analytics.